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Review

Applications of Artificial Intelligence in Fisheries: From Data to Decisions

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Abstract

AI enhances aquatic resource management by automating species detection, optimizing feed, forecasting water quality, protecting species interactions, and strengthening the detection of illegal, unreported, and unregulated fishing activities. However, these advancements are inconsistently employed, subject to domain shifts, limited by the availability of labeled data, and poorly benchmarked across operational contexts. Recent developments in technology and applications in fisheries genetics and monitoring, precision aquaculture, management, and sensing infrastructure are summarized in this paper. We studied automated species recognition, genomic trait inference, environmental DNA metabarcoding, acoustic analysis, and trait-based population modeling in fisheries genetics and monitoring. We used digital-twin frameworks for supervised learning in feed optimization, reinforcement learning for water quality control, vision-based welfare monitoring, and harvest forecasting in aquaculture. We explored automatic identification system trajectory analysis for illicit fishing detection, global effort mapping, electronic bycatch monitoring, protected species tracking, and multi-sensor vessel surveillance in fisheries management. Acoustic echogram automation, convolutional neural network-based fish detection, edge-computing architectures, and marine-domain foundation models are foundational developments in sensing infrastructure. Implementation challenges include performance degradation across habitat and seasonal transitions, insufficient standardized multi-region datasets for rare and protected taxa, inadequate incorporation of model uncertainty into management decisions, and structural inequalities in data access and technology adoption among smallholder producers. Standardized multi-region benchmarks with rare-taxa coverage, calibrated uncertainty quantification in assessment and control systems, domain-robust energy-efficient algorithms, and privacy-preserving data partnerships are our priorities. These integrated priorities enable transition from experimental prototypes to a reliable, collaborative infrastructure for sustainable wild capture and farmed aquatic systems.

Keywords: remote sensing; machine learning (ML); precision aquaculture; electronic monitoring (EM); IUU fishing detection; digital twins

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1. Introduction

Global fisheries and aquaculture are undergoing significant transformation due to intersecting pressures, including climate change, biodiversity loss, overfishing, and increasing demand. Traditional monitoring methods, such as trawl surveys, onboard observers, and manual taxonomy, are often expensive, limited in spatial and temporal coverage, and

prone to human bias [1–3]. Over one-third of estimated stocks are overexploited, and aquaculture now accounts for more than half of global seafood [1]. This field necessitates approaches that can be scaled across various habitats and fleets, integrate diverse evidence streams, and present uncertainty in decision-relevant formats. Recent advancements in artificial intelligence (AI), particularly in machine learning (ML), deep learning (DL), reinforcement learning (RL), and artificial intelligence of things (AIoT), have facilitated automated perception, short-horizon forecasting, and real-time control through the use of underwater imagery, acoustic echo-sounders, satellite sensors, and environmental DNA (eDNA) [4–6]. Instead of replacing established practices, these systems enhance their scope by expediting analysis, increasing observation density, and connecting measurements to operational actions.

Recent studies in fisheries science demonstrate that video pipelines and advanced detectors/transformers enhance underwater fish recognition despite turbidity and occlusion. Significant contributions consist of underwater videos [7], EM video for species classification [8], enhanced YOLO (You Only Look Once)-based underwater fish detection [9,10], the DeepFish benchmark [11] for complex habitats, and transformer-enhanced YOLO variants aimed at aquaculture [12,13]. Furthermore, transfer learning reduces the requirement for extensive annotations and enhances generalization across diverse habitats [14–17]. Environmental DNA (eDNA) metabarcoding serves as a sensitive, non-invasive method for assessing community composition, frequently aligning with traditional indices. Genomic techniques, including multi-marker and genome-editing strategies, offer enhanced resolution of stock structure, local adaptation, and aquaculture traits, though challenges remain regarding resource allocation and governance [18–22]. Precision aquaculture is experiencing significant advancements: supervised machine learning enhances rationing; reinforcement learning optimizes feeding and water quality in recirculating systems; and vision-based welfare and disease detection systems expand coverage at the farm level. Additionally, digital twins are proposed as a framework for harvest planning, but their implementation is currently theoretical in most operational contexts, requiring real-time bidirectional data flow and complex calibration that exceeds current farm data maturity [6,23–28]. In fisheries management, anomaly detection in Automatic Identification System (AIS) trajectories aids in identifying Illegal, Unreported, and Unregulated (IUU) fishing. Visible Infrared Imaging Radiometer Suite (VIIRS), known as satellite night lights, and Synthetic Aperture Radar (SAR) reveal "dark fleets," while enhanced monitoring through computer vision improves bycatch documentation. The integration of AIS/VMS with SAR/VIIRS, along with human-in-the-loop validation, connects algorithmic signals to enforcement outcomes [2,8,29–31]. In conjunction with these advancements, acoustic and multi-sensor analytics are progressing towards practical application. Weakly supervised Convolutional Neural Networks (CNNs) minimize the need for manual labeling of echograms. YOLO-style detectors enable near-real-time identification of fish schools in underwater video, achieving speeds exceeding real-time and demonstrating high accuracy in complex scenes [9,20,32,33]. U-Net-style segmentation enhances pixel-level accuracy in delineating features such as body and lamina boundaries, fins, and school extents, which are essential for size-to-mass conversions and other biomass proxies in aquatic monitoring [34–38]. Buoy-acoustic models, calibrated to set-by-set catches, are currently informing tuna fisheries [39–44].

Despite these advancements, significant limitations impede sustainable and equitable outcomes. Domain shift, influenced by turbidity, illumination, seasonal variation, and variations in gear and frequency, adversely affects performance on rare taxa and complicates transferability across sites and fleets [11,45–49]. There is a notable scarcity of standardized, multi-region datasets that encompass long-tail and protected species while linking imagery

with traits and environmental DNA [15,17,50–52]. Numerous studies present accuracy metrics without examining the propagation of errors to abundance indices, bycatch estimates, feeding and water quality control, or enforcement and compliance outcomes, thereby restricting decision relevance [40,42]. Emerging benchmarks for multi-frequency acoustics, electronic monitoring (EM) video, and fused AIS + SAR/VIIRS are characterized by limited standardization for acoustic targets, the nascent state of EM video datasets, and the scarcity of publicly available AIS–SAR/VIIRS fusion resources [53–57]. This includes assessments against observers, set-by-set catches, and boarding results, as well as multi-season, multi-site deployments that measure calibration drift, sensor fouling, missing data, computational and energy costs, and lifecycle maintenance requirements for edge devices [23,30,58,59]. Governance and equity issues, such as smallholder ROI, interoperability and cybersecurity, data-sharing incentives, and the integration of genomic insights into standard management practices, have yet to be fully resolved [18,60,61].

This review identifies existing gaps by synthesizing recent advances in fisheries science, particularly in genetics & monitoring, aquaculture, management, and sensing & technology. It proposes a research agenda to deploy effective, uncertainty-aware, and equitable artificial intelligence in the aquatic sector. We synthesize evidence across vision-based systems, bioacoustics, environmental DNA, automatic identification systems, satellite fusion, reinforcement learning, artificial intelligence of things, and genomic tools. Our analysis characterizes failure modes and traces how model uncertainty propagates across ecological, welfare, economic, and compliance outcomes. We identify critical benchmarking and data standards, emphasizing the need for multi-sensor, multi-frequency datasets that encompass rare taxa and employ auditable annotation protocols. We outline deployment frameworks that integrate domain-adaptive algorithms with digital-twin decision support systems and governance mechanisms that enhance participation by smallholder producers. Our overarching goal is to transition artificial intelligence into fisheries from experimental prototypes to functional infrastructure that enables sustainable, resilient, and transparent aquatic resource production.

This review systematically highlights both accomplishments and shortcomings. This study identifies failure modes, including performance degradation in rare and cryptic species, sensitivity to variations in turbidity and illumination, overfitting to specific farm conditions, and the widespread lack of formal uncertainty quantification in studies focused on operational applications. Instead of portraying artificial intelligence as a fully developed operational solution, we view it as a collection of promising yet evolving tools, with their practical value fundamentally dependent on data design, transparent evaluation standards, and adaptive governance structures.

This paper systematically integrates technical advancements with implementation requirements. Section 2 outlines the methodological framework by specifying the literature search protocol and the quality assessment criteria. Sections 3–6 delineate four primary application domains arranged hierarchically, progressing from raw data acquisition to operational decision support. Section 3 integrates fisheries genetics with monitoring technologies; Section 4 analyzes precision aquaculture systems; Section 5 evaluates fisheries governance and surveillance; and Section 6 discusses the fundamental sensing and computational infrastructure necessary for these applications. Section 7 addresses overarching challenges related to data governance, domain robustness, and equity considerations. Section 8 outlines the practical implications for policymakers, managers, farmers, and technology providers. Section 9 concludes by examining the transition from artificial intelligence prototypes to operational infrastructure. Figure 1 summarizes the overall conceptual framework and structures of this review.

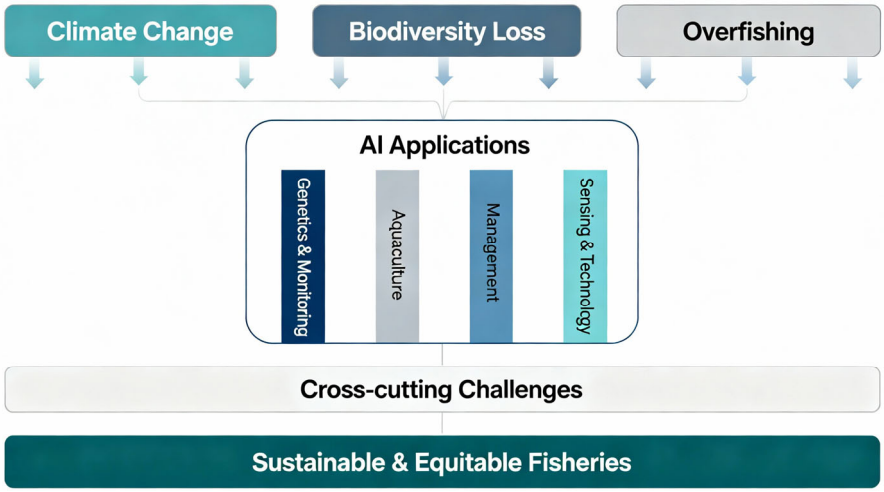

Figure 1. Conceptual Framework of the Review.

2. Literature Search Methodology

This paper provides a structured review of recent applications of AI in fisheries genetics, aquaculture, management, and enabling technologies. Inclusion is limited to peer-reviewed journal articles and conference papers published in English from January 2015 to June 2025, along with select technical reports that provide comprehensive methodology and evaluation protocols. Preprints were included only if they introduced novel architecture, such as foundation models, that had not been published in journals.

The subsequent keyword groups were employed to retrieve literature from interdisciplinary and domain-specific databases, including Web of Science Core Collection, Scopus, ScienceDirect, IEEE Xplore, and Google Scholar. (i) Domain terms: fisheries genetics, aquaculture, fisheries management, fisheries technology, sensing, monitoring; (ii) AI techniques: artificial intelligence, machine learning, deep learning; (iii) Application terms: species recognition, biomass estimation, stock assessment, bycatch, illegal, unreported, and unregulated (IUU) fishing, water quality forecasting, precision feeding, genomics, biofloc, artificial intelligence of things (AIoT). Search terms were tailored for each database (e.g., "aquaculture" AND ("deep learning" OR "machine learning") AND ("feeding" OR "water quality"). Additionally, forward–backward snowballing was employed to identify pertinent studies from reference lists and citation networks. The studies were evaluated based on the clarity of their data sources, the robustness of their validation methods (e.g., independent test sets compared to random splits), and the reproducibility of their reported metrics.

Eligibility criteria for studies included: (i) the application of AI or machine-learning methods to fisheries or aquaculture issues, (ii) provision of fundamental data regarding sampling design, sample size, and context, and (iii) reporting of quantitative performance metrics related to prediction, classification, detection, control, or decision support. Conceptual studies, hardware studies lacking AI, and performance claims devoid of methodological explanation were excluded.

3. AI in Fisheries Genetics and Monitoring

Recent advancements in computer vision, acoustics, environmental DNA, and genomics are transforming the monitoring and evaluation of wild fish populations. This section investigates the evolution from individual detection tasks in underwater imagery to comprehensive molecular and multi-sensor methodologies for ecosystem-scale assessment frameworks. This study initiates with the detection of species in challenging environmental conditions, explores transfer learning strategies that generalize across various habitats and seasons, and examines image enhancement techniques for turbid or degraded data. We

subsequently integrate environmental DNA metabarcoding, acoustic inference, and multimodal fusion for the assessment of community-level dynamics. This section concludes by analyzing how genomic analysis and trait-based modeling improve understanding of population dynamics and inference of stock structure. This conceptual framework illustrates the integration of visual, acoustic, and genomic signals, facilitating continuous non-invasive monitoring systems that can supplant infrequent, observer-dependent surveys (Table 1).

3.1. Species Detection and Classification

Contemporary convolutional backbones and efficient detectors have significantly enhanced automated species recognition [12,14,17,62]. Nonetheless, performance diminishes in response to habitat shifts (e.g., turbidity, illumination, seasonality) and for long-tail or protected taxa like elasmobranchs, which has consequences for stock assessments and ecosystem models [15,17,50]. Underwater-tuned architectures that enhance local–global representations, combined with stereo imaging to stabilize tracks and minimize duplication in turbid water, improve robustness [63]. Advancement relies on comprehensive, standardized image-trait datasets, encompassing bycatch and discard categories, alongside 3D-aware perception to address coverage deficiencies and facilitate trait-informed evaluations [50].

Gains in species recognition should be analyzed in light of the underlying datasets. Research that optimizes ResNet-50 or EfficientNet on small, curated datasets (e.g., 500–600 images across 20 species or a few thousand market images covering 10–20 taxa) tends to report high accuracy levels. However, such studies offer limited insights into robustness with respect to rare taxa, life-stage variation, or domain shifts across different laboratories, farms, and wild habitats [5,64,65]. In contrast, benchmarks like DeepFish, which include approximately 40,000 high-resolution images from 20 habitats, subject detectors to more realistic variations in habitat and illumination. This exposure highlights systematic performance degradation for small, distant, and partially occluded fish, as well as during transitions between habitat types [11,66]. Future research must consistently report dataset size, class imbalance, and habitat coverage alongside overall accuracy metrics, clearly distinguishing performance on small, curated datasets from that on extensive, in situ benchmarks.

3.2. Transfer Learning and Domain Adaptation

Fine-tuning pre-trained Convolutional Neural Networks (CNNs) decreases the need for extensive labeling in novel habitats; however, domain-specific biases, such as color cast and backscatter, negatively impact recall rates for rare species [11,14,67–70]. Research supports the development of underwater-specific detector designs and video-first pipelines that ensure temporal consistency, enhancing generalization across various sites and seasons [15,17,50]. Recent developments in stereo–video and multi-fish tracking datasets [71,72] support active learning and disagreement-driven annotation [73,74]. Integrating perception with SLAM/NeRF-style geometry [75–77] offers priors that enhance cross-domain transfer in extended deployments. Extensive, multi-regional training datasets that specifically include protected and low-abundance taxa are crucial for reducing structural bias in monitoring [50].

3.3. Low-Resolution and Noisy Image Handling

Numerous underwater datasets exhibit low resolution and are compromised by backscatter [11,78,79]. Generic super-resolution enhances perceived detail but poses risks of hallucinations that may distort ecological endpoints, such as counts and lengths [80–82]. Task-aware pipelines that integrate enhanced super-resolution methods, such as Enhanced Super-resolution Generative Adversarial Networks (ESRGAN), with downstream detectors and validate against native-resolution truth have successfully restored recognition perfor-

mance while maintaining low false positive rates in the context of uncertainty reported for super-resolution-assisted detections [83,84]. Method development must emphasize calibration and error quantification within realistic turbidity conditions to guarantee suitability for management purposes.

3.4. Environmental DNA (eDNA) Monitoring

Environmental DNA (eDNA) metabarcoding serves as a sensitive, non-invasive method for assessing community composition and seasonal turnover, effectively complementing traditional gear-based surveys. Numerous studies have demonstrated its alignment with conventional indices across various systems [3,85–87]. Significant challenges involve correlating read counts with abundance, considering hydrological transport, and tackling incomplete reference libraries. Current guidance highlights the importance of standardized controls, such as qPCR/dPCR and field blanks, the co-design of acoustic and video indices, and the explicit incorporation of detection and transport uncertainty into assessment frameworks [88]. The targeted curation of reference libraries for understudied faunas will enhance reliability and comparability.

3.5. Acoustic Inference and Multi-Sensor Fusion

Machine-learning pipelines for fisheries acoustics are evolving towards survey-grade integration, incorporating methods to embed acoustic classifiers within estimation procedures [89]. Cabled observatories demonstrate that synchronized video and acoustic streams yield complementary biodiversity signals; however, they underscore the necessity for more extensive labeled acoustic corpora and systematic cross-modal analysis [90]. The coordinated and time-aligned integration of acoustics, imaging, and—where applicable—eDNA, along with uncertainty outputs, provides a practical approach to operational monitoring and assessment at the ecosystem level.

3.6. Genomics and Breeding Technologies

Population genomics and genome editing are progressing swiftly; however, genomic resources are not uniformly distributed among commercially and ecologically significant species, and the adoption of routine management practices is constrained [18,91]. Multi-marker and multi-omics approaches demonstrate potential for elucidating stock structure, local adaptation, and connectivity [18,92–97]. Genome editing techniques, such as CRISPR/Cas, are being investigated for enhancing disease resistance and production traits in aquaculture, presenting both technical opportunities and governance challenges [19,98]. Key priorities encompass the establishment of reference genomes for essential species, the development of standardized data protocols, and the enhancement of collaboration among geneticists, managers, and policymakers to promote the responsible application in both wild and cultured stocks.

3.7. Trait-Based and Statistical Modeling Approaches

Trait-based indicators and hierarchical/mixed-effects models are increasingly employed to connect life-history traits (such as growth, maturation, and longevity) to fishing pressure and climate variability, effectively capturing heterogeneity at both individual and population levels. Recent applications of Generalized Linear Mixed Models (GLMM) and Linear Mixed-Effects Models (LMEM) indicate that community trait composition correlates with fishing intensity, even when controlling for environmental covariates [99–101]. Nonetheless, the inconsistency of trait databases across taxa and regions limits their general applicability [102]. Enhancing trait coverage, such as fecundity, natural mortality, and maturation schedules, along with incorporating mixed-effects methods into assessment workflows, can enhance inference and forecasting. This improvement is particularly notable when combined with genetic and remote-sensing indicators [102,103].

4. AI in Aquaculture

Artificial intelligence is facilitating a shift in aquaculture from reactive to adaptive production systems that optimize the competing objectives of yield, welfare, profitability, and environmental impact amid uncertainty. This section integrates AI applications throughout the aquaculture farm management process. The initial focus is on real-time operational control, employing supervised learning for feed rationing and conversion efficiency, reinforcement learning for optimizing water quality in recirculating systems, and vision-based monitoring for welfare and disease assessment. This study analyzes forecasting tools for dissolved oxygen, ammonia, welfare outcomes, and harvest timing, integrating growth ensembles and market scenarios. This section addresses emerging capabilities such as climate scenario simulation, explainable AI for farmer adoption, biofloc optimization, and Artificial Intelligence of Things (AIoT) infrastructure. This work highlights significant discrepancies between offline validation and operational value, as well as the importance of uncertainty quantification and manager-in-the-loop testing for transitioning from prototypes to decision-grade systems (Table 2).

4.1. Automated Feeding Systems

AI-enabled feeders that integrate computer vision with sequence models, such as Convolutional Neural Networks (CNN) and Gated Recurrent Units (GRU), can accurately predict short-term feed demand and growth, thereby minimizing waste and enhancing feed conversion efficiency [6,104–106]. Most evaluations, however, are short-term, species-specific, and conducted in tank environments, which restricts their applicability to net-pens and mixed cohorts [6,55]. A practical approach involves vision-in-the-loop growth estimation integrated with explainable, auto-recalibrating temporal models (Figure 2) that adjust to biomass and seasonal variations, while real-time water quality forecasts limit rationing to prevent post-prandial hypoxia [6,27]. Future trials ought to be prospective, multi-site, and multi-season, incorporating standardized reporting on calibration drift, turbidity/low-light failure modes, and whole-farm cost–benefit [107,108].

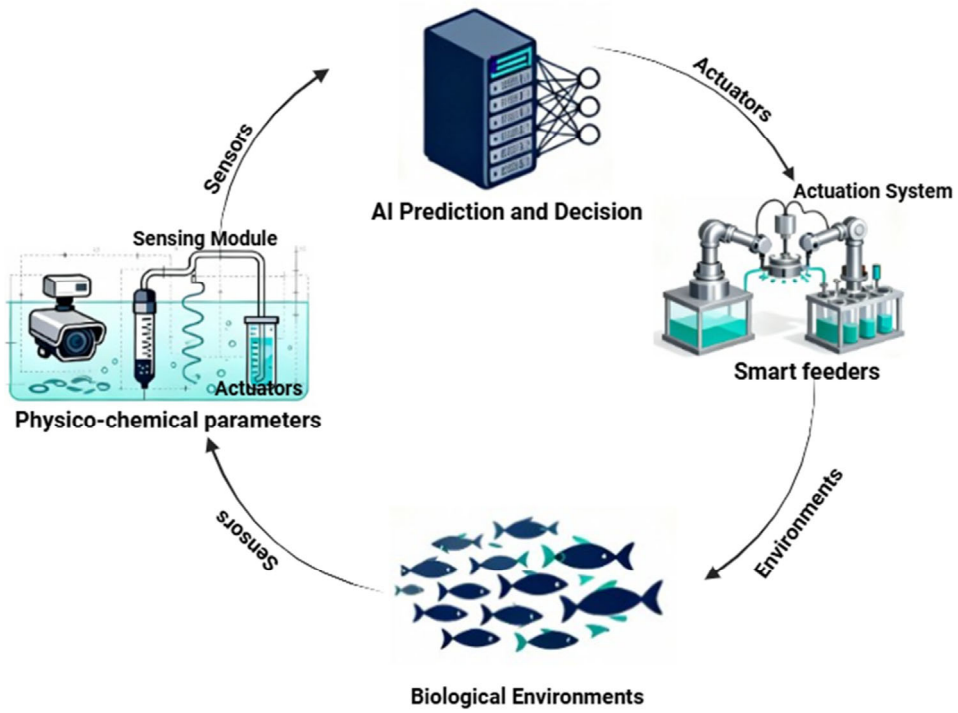

Figure 2. AI-Enabled Aquaculture Control Loop.

4.2. Machine Learning for Feeding Optimization

Supervised machine-learning techniques, including gradient boosting, support vector regression, and elastic-net ensembles, enhance rationing by integrating appetite cues, growth history, and environmental covariates [25,109,110]. Models frequently exhibit overfitting to single-farm telemetry and demonstrate reduced performance when there are changes in sensor coverage, density, or feed formulations [25,107]. Deployments must implement feature governance (sensor quality control, imputation), conduct domain-shift testing with regular retraining, and utilize uncertainty-aware throttling to reduce ratios when confidence decreases [25,108]. Key priorities encompass the development of shared, anonymized multi-farm datasets and benchmarks that evaluate environmental externalities, such as nitrogen discharge, in conjunction with accuracy and profitability [111].

4.3. Reinforcement Learning Controllers

In recirculating aquaculture systems, reinforcement learning, an intense deterministic policy gradient, has co-optimized feeding and water quality control amidst nonlinear dynamics [24,112]. Nonetheless, the training process requires substantial data, the design of rewards is sensitive, and the safeguards for infrequent catastrophic events, such as aeration failure, are inadequately defined [24,113]. The integration of offline reinforcement learning pretraining using Supervisory Control and Data Acquisition (SCADA) logs, alongside action-clipping, stringent safety regulations, and physics-informed surrogates, enhances sample efficiency and safety. Additionally, joint energy optimization presents a further advantage [24,112]. Conducting comparative studies against robust Model Predictive Control (MPC) baselines on identical Recirculating Aquaculture Systems (RAS) lines, utilizing longitudinal welfare and energy metrics, represents a crucial subsequent step [108,114].

4.4. Water Quality Forecasting

Deep learning applied to dense IoT streams facilitates hours-ahead forecasting for dissolved oxygen, pH, and ammonia, thereby enabling proactive aeration and resource management [45,100,115]. Vulnerabilities encompass sensor fouling, data absence, and concept drift as biomass and temperature regimes change [27,107]. Effective systems ought to integrate attention-based or hybrid architectures with automated gap-filling and drift detection, while also incorporating forecast uncertainty into control logic. Decomposition-assisted CNN-RNN (Recurrent Neural Networks)/Transformer stacks show potential [27,116]. Multi-season validations that concurrently predict dissolved oxygen and nitrification products will diminish "false-safe" signals and enhance reliability [23].

4.5. Welfare and Stress Monitoring

Vision pipelines are transitioning from biomass and length estimation to behavior-based welfare indices, such as erratic swimming and clustering [117–119]. However, generalization is impeded by factors such as poor visibility, crowding, and sparse event labels from commercial farms [26,120]. The integration of passive video, multi-object tracking, and low-cost environmental sensing, can enhance reliability. Disease-focused detectors utilizing robust backbones, such as YOLO variants, demonstrate lesion and disease detection rates exceeding 90% under rigorous conditions, although they require substantial labeling efforts [111,121]. Future work should be anchored in community-curated, expert-annotated welfare corpora and intervention-focused evaluations, specifically targeting mortality reduction and welfare gains [107].

4.6. Harvest Forecasting and Planning

Time-series artificial intelligence that integrates growth histories with feeding and environmental data can predict optimal harvest periods and synchronize logistics with market quality [122]. However, existing studies predominantly focus on retrospective analyses and seldom quantify decision uncertainty [6,123,124]. Incorporating growth ensembles into digital-twin dashboards that display confidence intervals and simulate alternative harvest dates based on weather and price scenarios, while being continuously recalibrated to grading data, can enhance profitability and mitigate risk [6,26]. Manager-in-the-loop trials that are prospective in nature, comparing AI-guided schedules with heuristics, are necessary to determine operational value [108].

4.7. AI Models for Climate Scenario Simulations

Emerging generative AI models for climate scenario simulation currently lack empirical validation. The likelihood of plausible yet mechanistically unfounded hallucinated scenarios is significant. Hybrid approaches that integrate generative components with physics-informed constraints demonstrate greater potential, yet they remain in the experimental phase. Generative AI is being utilized to evaluate farms under scenarios including heat waves, hypoxia, and pathogen introductions [125]. Although simulators show potential for enhancing preparedness, they encounter validation deficiencies and risks associated with biased prior assumptions [125,126]. Hybrid approaches that integrate generative components with mechanistic bioenergetics and epidemiological submodels, while validating realism through historical extremes and expert elicitation, present a viable advancement [125]. Open benchmarks that assess scenario fidelity and decision usefulness, such as avoided loss and response time, will facilitate responsible adoption [108].

4.8. Explainable AI (XAI) for Farmer Trust

To facilitate the adoption of AI systems in commercial aquaculture, their decision-making processes must be transparent and interpretable for operators. Models characterized as "black box" demonstrate high accuracy; however, they often struggle to build trust when suggesting essential interventions, such as disease treatment or harvest timing. Explainable AI (XAI) techniques, including SHAP (Shapley Additive Explanations) and Grad-CAM, are increasingly used to visualize the features that influence model predictions, such as identifying specific skin lesions or swimming behaviors that trigger a disease alert [127,128]. XAI tools offer visual justification for automated decisions, facilitating "human-in-the-loop" verification. This allows farmers to confidently assess AI-driven recommendations prior to implementing significant operational changes [129,130].

4.9. Biofloc System Optimization

Biofloc technology (BFT) effectively recycles nutrients and stabilizes water quality; however, operators face challenges in managing microbial dynamics in relation to carbon-to-nitrogen ratios, aeration, and temperature [131,132]. The integration of continuous sensing with predictive models for floc volume, nitrification rates, and sludge removal, along with alarms for off-nominal microbiome states, diminishes the burden on operators. Standardized multispecies BFT datasets, including endpoints such as FCR, TAN spikes, and off-flavor risk, are essential for comparing AI guidance with rule-based management [107].

4.10. Smart Aquaculture with AIoT

The vision of the 'Internet of Things' presupposes ubiquitous, dependable connectivity and consistent energy availability. Vessel networks operate in remote regions with intermittent satellite connectivity and limited power, which limit real-time processing capabilities.

The AIoT stack, comprising networked sensors and edge/cloud analytics, currently supports "smart farms" that incorporate automated feeding, water quality management, and disease notification systems [6,61,133]. Challenges in adoption remain concerning interoperability, cybersecurity, connectivity, and return on investment for smallholders [23,60]. Tiered architectures, including edge inference and cloud retraining, alongside standards-based telemetry and lifecycle cost modeling, enhance reliability and economic efficiency. Recent reviews provide a catalog of scalable reference designs and considerations for data and AI governance [61,107]. Prioritizing open testbeds and multi-stakeholder pilots is essential for mitigating investment risks and validating performance in commercial settings.

5. AI in Fisheries Management

Modern fisheries governance relies on risk-based management that integrates various sensor data with machine learning and human oversight to produce decision-grade indicators. This section investigates AI applications throughout the governance value chain, structured from vessel-level surveillance to ecosystem-scale integration. This study focuses on vessel-centric anomaly detection within automatic identification system trajectories to identify illegal, unreported, and unregulated fishing. It also involves mapping global fishing effort through multi-sensor fusion to address gaps in small-scale fleet coverage, as well as using satellite radar and nighttime imagery to identify non-broadcasting vessels. The discussion includes programmatic tools such as electronic monitoring video for documenting catch and bycatch, standardizing interactions with protected species, assessing transshipment risks, and detecting vessels across sensors in marine protected areas. The framework underscores the need to validate algorithmic outputs against enforcement outcomes, including boardings, port inspections, and observer records. Furthermore, it asserts that recommendations should be incorporated into transparent decision-making frameworks that have clearly defined objectives and risk tolerances (Table 3).

5.1. Illegal, Unreported, and Unregulated (IUU) Fishing Detection

Machine-learning analyses of AIS trajectories increasingly identify behavioral anomalies such as course and speed shifts, boundary loitering, and suspicious silences, which assist in prioritizing enforcement efforts toward higher-risk events [30,31,134]. However, the effectiveness of these analyses depends on the completeness of signals and the accuracy of verified labels. Additionally, issues such as AIS spoofing and disablement, along with low regional uptake, present significant challenges [29,30]. A practical approach involves multi-source fusion, integrating AIS with satellite detections (SAR, VIIRS) and VMS, alongside anomaly typologies that specifically address identity manipulation and silence. This should be complemented by human-in-the-loop feedback from boardings to ensure alignment of models with enforcement outcomes [29,30].

5.2. Global Fishing Effort Mapping

Global AIS-based maps indicate that industrial fishing occurs in over half of the ocean, significantly altering the evidence base for governance [2,57]. However, these products inadequately represent artisanal and small-scale fleets, as well as areas with limited AIS coverage, leading to biased assessments of footprint and equity [2]. Mitigation necessitates the integration of AIS with complementary sensors such as VIIRS nighttime lights and SAR, alongside administrative sources like VMS and registries. Additionally, it is essential to provide uncertainty layers that indicate areas where efforts may be underreported [2,58]. Key priorities encompass the establishment of sustained multi-sensor pipelines and the implementation of periodic re-estimation to quantify changes driven by policy, economics, and climate factors.

5.3. Electronic Monitoring and Bycatch Detection

Computer vision models utilized in EM video can effectively identify fishing events, quantify catch, and detect bycatch taxa, thereby enhancing coverage and decreasing dependence on human observers [135–137]. Nonetheless, challenges such as label scarcity, occlusions, variable camera placement, and mixed catches hinder generalization and species-level accuracy [8]. Programs must implement active learning to address challenging frames, facilitate transfer learning among fleets, and establish standardized annotation and audit protocols that combine algorithmic outputs with swift expert adjudication for regulatory purposes [8]. Future research should conduct prospective, multi-fleet trials that directly compare EM-AI indicators with observer and port-sampling endpoints, such as bycatch rates, rather than relying solely on computer vision scores. A further limitation is that many EM studies rely on proprietary datasets drawn from a small number of fleets or regions, with limited disclosure of sampling protocols, annotation consistency, and the long-term maintenance of labeling standards. This constrains external reproducibility and makes it difficult to assess how well published detectors would generalize to fleets that differ in gear, lighting, and catch composition.

5.4. Protected Species Oversight

Electronic monitoring combined with artificial intelligence (EM + AI) has the potential to standardize the documentation of interactions with protected species, such as turtles, seabirds, and marine mammals, across various vessels and gear types [138,139]. This approach frequently matches or surpasses the effectiveness of traditional observer coverage in terms of effort and event detection [140–142]. Performance for rare or cryptic taxa is limited by class imbalance and insufficient training sets, necessitating expert validation and rigorous QA/QC to minimize false positives and negatives [142,143]. Research on workflow indicates that optimized review protocols and triage can alleviate analyst workload while maintaining detection quality; however, costs and backlogs may shift onshore without meticulous process design [144]. Future research will prioritize the curation of open, labeled corpora that encompass gears, lighting, and rare-event edge cases; the integration of human-in-the-loop auditing with calibrated error metrics for management thresholds; and the expansion of pilots in high-risk fisheries utilizing standardized camera placement and metadata to facilitate transfer learning and the development of reliable protected-species classifiers [142–144].

5.5. Dark Fleet Detection

Nighttime lights (VIIRS) and satellite radar (SAR) effectively identify non-broadcasting "dark" vessels, addressing gaps in AIS data [31,57,145]. However, detection is influenced by weather and moonlight conditions, and associating detections with specific identities or activities poses significant challenges [58,145,146]. Optimal methodology involves cross-sensor fusion that integrates VIIRS and SAR data in a space–time framework, utilizing available AIS/VMS information and employing probabilistic matching validated through targeted patrols [58,145,146].

5.6. AI-Enhanced Stock Assessment and Decision Support

The shift from data-intensive monitoring to decision-quality recommendations requires integrating AI with population dynamics models. Conventional stock assessments often rely on oversimplified assumptions, potentially neglecting complex environmental factors. Prediction skill metrics are utilized to validate integrated assessment models within ensemble frameworks, enabling managers to objectively evaluate or dismiss model hypotheses based on their predictive performance on unseen data [147]. Emerging ad-

vanced decision support systems (DSS) leverage real-time data streams to enhance fishing routes and gear selectivity, thereby reducing fuel consumption and bycatch mortality while optimizing economic returns [148].

5.7. Transshipment Risk Assessment

AIS trajectory analysis can detect at-sea rendezvous between fishing vessels and reefers, thereby enhancing inspection targeting to combat the laundering of illicit catch [149,150]. However, false positives may occur due to benign proximity and incomplete AIS histories [149]. Risk scoring must integrate movement features, such as speed, headings, and duration, with contextual priors, including flag and ownership histories, port states, and prior behavior. This approach should be audited against boarding outcomes to refine thresholds [149]. Future analyses must incorporate SAR/VIIRS to identify non-broadcasting participants and evaluate the impact of risk-based inspections on compliance behavior over time.

5.8. Cross-Sensor Fusion for Vessel Detection

The integration of AIS with optical and SAR imagery enhances the detection of small vessels in Marine Protected Areas (MPAs) and Exclusive Economic Zones (EEZs) [151–153]. However, challenges such as reliance on cloud cover for optical data, limitations in revisit frequency, and computational demands hinder continuous monitoring [153–155]. Deployments must utilize modular pipelines that default to SAR in cloud environments, incorporate VIIRS for nighttime detection, and implement uncertainty-aware tracking to preserve identity across sensors while identifying incursions [153–155].

6. AI in Sensing and Fisheries Technology

The effectiveness of artificial intelligence in fisheries and aquaculture is fundamentally contingent upon the maturity and reliability of sensing infrastructure and computational frameworks. This section analyzes developments in acoustic interpretation, edge computing, and foundation models that establish critical data foundations for operational applications. This study focuses on automating acoustic echo-trace analysis to reduce manual annotation in multispecies surveys. This study examines real-time fish detection using YOLO-based methods for identifying schools in echograms and underwater video. It also addresses integrating detection outputs into survey-grade abundance estimation while accounting for model uncertainty. This study investigates lightweight architectures for edge deployment on power-constrained platforms, including solar-powered buoys and battery-operated systems. This section addresses foundation models, encompassing large language models tailored for marine domains, vision–language models designed for open-vocabulary underwater recognition, and vision–language action systems applicable to multi-robot aquaculture operations (Table 4).

6.1. Acoustic Echo-Trace Classification

Weakly supervised convolutional neural networks (CNNs) trained on minimally annotated echograms can automate the labeling of echo traces, thereby decreasing the effort required from experts in multispecies pelagic environments [40]. Significant bottlenecks consist of the scarcity of standardized multi-frequency corpora and domain shifts across gears, frequencies, and regions, as well as the ambiguous propagation of classification errors into biomass indices [51,52]. Key priorities include the establishment of open benchmark datasets featuring harmonized protocols and comprehensive metadata, as well as the implementation of transfer and self-supervised pretraining to enhance generalization [40,52]. Future research should quantify the comprehensive effects on abundance and biomass, as well as illustrate cross-basin transfer.

6.2. YOLO-Based Fish Detection in Echograms

Modifications of YOLOv5 for commercial echo-sounder data facilitate accurate identification of individual fish and schools and have also aided in instrument calibration, thus allowing for near-real-time operational application [156]. Practical constraints encompass data intensity, sensitivity to absent pings or uncalibrated instruments, and computational demands on small platforms [156]. Remedies encompass physics-aware augmentation (e.g., simulated noise/missing lines), multi-frequency inputs, temporal tracking, and edge inference. Evaluations must report detection-to-assessment utility, rather than relying solely on mean Average Precision (mAP) [52]. The preference for YOLO variants in fisheries applications is primarily due to their high inference speed, which facilitates real-time edge deployment on power-constrained vessels, despite known limitations with small, occluded targets in dense schools. Recent Transformer-based architectures, such as the Swin Transformer, enhance global contextual reasoning and occlusion handling. However, their significantly increased latency on edge hardware limits routine operational use. This situation underscores the importance of training protocols, especially physics-informed augmentation techniques (e.g., turbidity, backscatter, partial occlusion) and robust regularization, over the choice of architecture itself [9,34]. Single-stage YOLO detectors utilizing shared backbones and multi-scale heads can achieve real-time inference rates of 30 to 100 frames per second on standard GPUs while maintaining competitive mean Average Precision (mAP) on extensive fish datasets. Architectural enhancements, including spatial pyramid pooling, re-parameterized convolutions, attention-based necks, and task-specific loss functions, typically result in incremental improvements in localization and small-object recall under specific underwater conditions [32,63,156–158]. Nonetheless, these improvements are generally shown on individual datasets and are infrequently assessed under systematic domain shifts (e.g., clear versus turbid water, varying gear types), which restricts confidence in their applicability beyond benchmark contexts. The selection between standard YOLO, Transformer-enhanced variants, and multi-head hybrids must be clearly defined in the context of deployment constraints, including speed at target resolution on designated hardware, accuracy with uncertainty estimates, model size and computational load (parameters, FLOPs, quantization), and robustness under domain shift evaluated on held-out seasons, locations, or vessels, rather than approached as a theoretical model-selection issue. In numerous operational fisheries monitoring systems, the slight accuracy enhancements provided by more complex Transformer-enhanced YOLO models may be surpassed by the benefits of simpler architectures, which can be more clearly characterized, monitored, and maintained over time in the context of data drift and hardware constraints [159].

6.3. From Detection to Abundance Estimation

A significant challenge to implementing AI in fisheries surveys is integrating machine-learning results into existing abundance estimation processes. High precision, such as F1 scores, is essential; however, it is inadequate for management decisions if biases in classification affect biomass indices. Recent frameworks illustrate the integration of deep learning-based acoustic target classification (ATC) into survey estimation processes [89]. By comparing AI-derived estimates with traditional manual assessments in North Sea sandeel surveys, it was determined that model predictions should be evaluated based on their influence on the final survey estimate, rather than solely on image-level metrics [89]. The implementation of "estimation-aware" validation is crucial for maintaining time-series consistency in automated pipelines, which is necessary for stock assessment and quota determination.

6.4. Near-Real-Time Acoustic Monitoring

Solar-powered autonomous echo-sounders, when integrated with machine-learning pipelines, are capable of streaming and summarizing biomass indicators with an hour-scale latency, thereby addressing limitations of traditional surveys [160]. Environmental reliability issues persist, including diel migrations and flow noise, alongside limitations in power and telemetry, as well as a lack of extensive long-term validation [59,160]. Proposed solutions encompass duty-cycled sampling, on-edge compression and inference, standardized quality control flags, and open dashboards. Relevant aquatic telemetry indicates that edge computing significantly reduces energy consumption while facilitating real-time analytics [161]. Future research should link alerts to management actions prospectively and evaluate seasonal robustness across various sites.

6.5. Tuna Biomass Estimation (TUN-AI)

TUN-AI integrates buoy echo-sounder data with gradient-boosting models and oceanographic principles to forecast biomass under drifting Fish Aggregating Devices (dFADs) at basin scales, thereby aiding decision-making [42]. Identified gaps encompass ambiguity in species composition and variations in buoy and frequency; it is imperative to calibrate and validate against set-by-set catches [42]. The integration of species-specific target-strength priors and recent TS measurements for tropical tunas enhances estimates and facilitates species discrimination [39,43]. Future study should include uncertainty measures, such as prediction intervals, and evaluate cross-ocean generalization.

6.6. Fish Sound Detection via Passive Acoustics

Machine learning and deep learning techniques are progressively utilized to detect and classify fish sounds and behaviors, facilitating scalable biodiversity monitoring [162]. Significant challenges include background noise, overlapping sources, and a scarcity of labeled libraries containing contextual metadata. Comparative analyses of classical and deep detectors highlight the necessity for reliable training data and uniform protocols [163,164]. Immediate priorities encompass the development of larger open annotated corpora, the enhancement of noise-robust front ends, and the implementation of weak/self-supervised learning. Future validations should connect acoustic activity to abundance/state and evaluate management value [162,164].

6.7. Lightweight Object Detection Models

Lightweight detectors designed for small echogram targets, such as YOLOv8-SBE, offer advantageous accuracy-latency trade-offs with fewer parameters, facilitating onboard and edge deployment [158]. Cross-frequency and cross-basin robustness, as well as noise-regime performance, are still insufficiently documented. Recommended practices involve pruning and quantization alongside explicit latency and energy reporting, as well as coupling detection for temporal tracking. Insights from related aquatic sensing suggest that significant power savings are achievable [161].

6.8. Foundation Models: Multimodal and Embodied AI Systems

The trajectory of AI in aquatic sciences is shifting from narrowly specialized, supervised architectures toward general-purpose Foundation Models characterized by multimodal reasoning, zero-shot generalization, and semantic understanding. This represents a fundamental departure from the task-specific paradigm that has dominated prior sections.

6.8.1. Large Language Models (LLMs)

Large language models designed for marine domains, including OceanGPT for ocean science and AQUA for aquaculture and fisheries, illustrate that domain-specific instruction

tuning and retrieval-augmented generation can facilitate literature synthesis, protocol summarization, and scenario exploration for researchers, regulators, and farmers [165,166]. Recent evaluations of artificial intelligence in aquaculture suggest that these technologies can enhance access to expertise while also revealing ongoing challenges, such as disparities in data and infrastructure, insufficient external validation, and ambiguous origins of recommendations, particularly for small-scale operators [107,167,168]. Research in medical imaging indicates that implementing generative models in safety-critical environments requires meticulously documented training and evaluation data, clear reporting of limitations, and established governance frameworks for human oversight [169]. In fisheries and aquaculture, it is essential to integrate domain-specific LLMs into clearly defined advisory workflows that include traceable sources, outputs that account for uncertainty, and regular auditing. This approach is preferable to employing them as generic conversational tools, where recommendations may be implemented without external verification.

6.8.2. Vision–Language Models (VLMs)

Vision–language models enhance these capabilities by aligning visual and textual representations across image and video streams. Generic models like Contrastive Language–Image Pretraining (CLIP) have demonstrated capabilities in open-vocabulary recognition and zero-shot transfer; however, their effectiveness in underwater scenes is limited by the restricted and biased representation of aquatic environments in their pretraining datasets [170]. Marine-specific foundation models, such as MarineInst and AquaticCLIP, fill this gap by utilizing extensive multimodal datasets comprising benthic, pelagic, and industrial imagery. These models demonstrate significant improvements in instance-level comprehension and adaptable querying for marine ecology and operations [171,172]. Most evaluations continue to depend on curated benchmarks characterized by controlled turbidity, lighting, and biofouling. There is a lack of systematic evidence on the performance of vision–language models in calibration, recall, and robustness as sensor conditions deteriorate, visibility varies, and species assemblages shift over extended deployments.

6.8.3. Vision–Language–Action (VLA) Models

Vision–language–action models integrate perception and language for control, enabling autonomous vehicles to comprehend high-level natural-language instructions and produce low-level actuation commands. Recent underwater frameworks, including Underwater System Integration and Management (USIM) with the U0 controller and UnderwaterVLA, demonstrate that VLA architectures can facilitate multi-task navigation, inspection, and manipulation by integrating multimodal fusion with task-level reasoning, all while operating under realistic hydrodynamic and communication constraints [76,173]. AquaChat and AquaChat++ illustrate the application of LLM-guided planning in aquaculture for coordinating multiple remotely operated vehicles. This includes adaptive net pen inspection, battery-aware mission planning, and fault-tolerant control [130,174]. These systems are primarily in the prototype phase, with validation limited to simulations or brief campaigns in controlled environments. Their resilience to long-term changes in infrastructure, environmental conditions, and operational practices has yet to be determined. A critical priority is to advance from proof-of-concept demonstrations to longitudinal comparisons with current operating procedures, evaluating safety, cost, reliability, and operator workload before VLA-based autonomy can be considered a mature infrastructure for fisheries and aquaculture.

6.9. Semantic Segmentation of Echograms

U-Net and DeepLab-style models effectively delineate discrete scatterers and aggregations, enhancing density and biomass estimation in comparison to heuristic thresh-

olding [44]. Outstanding challenges encompass elevated annotation costs, variability among annotators, class imbalance, and sensitivity to frequency-dependent appearance. Consensus-label workflows incorporating agreement metrics, semi-/self-supervised training, and data preparation strategies significantly influence ATC performance [51,52]. Recent advancements in segmentation within acoustic and sonar applications facilitate multi-task pipelines and the detection of surface and bubble boundaries, which may apply to fisheries echograms [175]. However, evaluations of U-Net and DeepLab variants in fisheries acoustics remain dominated by single summary statistics, typically IoU or F1 computed on limited sets of expert-annotated transects, metrics that are highly sensitive to annotation protocols, class imbalance, and frequency-dependent appearance and that only indirectly capture uncertainty in biomass or density estimates derived from segmentation masks [51,52]. Comparative analyses indicate that inter-annotator disagreement and seemingly minor post-processing choices (e.g., smoothing, morphological filtering) can materially alter biomass estimates even when headline segmentation scores remain similar [44]. Accordingly, segmentation pipelines should be appraised within an end-to-end inferential framework that explicitly links pixelwise predictions to downstream ecological indicators, with uncertainty propagated to abundance indices and management advice, rather than being judged solely on pixel-level performance metrics.

7. Cross-Cutting Challenges

Structural barriers across all application domains hinder the transition from laboratory-validated artificial intelligence prototypes to dependable operational systems. This section consolidates five categories of challenges that need to be addressed for operational deployment. We first analyze data composition and benchmarking standards, highlighting the prevalence of small curated datasets with inflated accuracy metrics that are inadequate under domain shift, the biased representation of commercially valuable species, and the limited availability of public benchmarks with comprehensive metadata. We analyze model architectures and training protocols, highlighting that algorithm selection is influenced by practical constraints such as latency, power, and maintenance, rather than solely by marginal improvements in accuracy. Additionally, we note that training decisions often have a greater impact on performance than the architecture's novelty. Third, we investigate system-level design considerations, such as digital twins, artificial intelligence of things platforms, and data-sharing collaboratives, which are still in early stages and require well-developed governance, maintenance, and business models. Fourth, we explore metrics and uncertainty quantification, emphasizing the common practice of reporting point accuracy without accompanying confidence intervals or propagation to assessment endpoints. We identify an operational gap: the need for prospective manager-in-the-loop validation within formal decision frameworks that connect algorithmic outputs to quantifiable improvements in stock status, welfare, profitability, and regulatory compliance. These challenges delineate the research agenda required for artificial intelligence to evolve from a diagnostic tool to a dependable framework (Figure 3) for aquatic resource management.

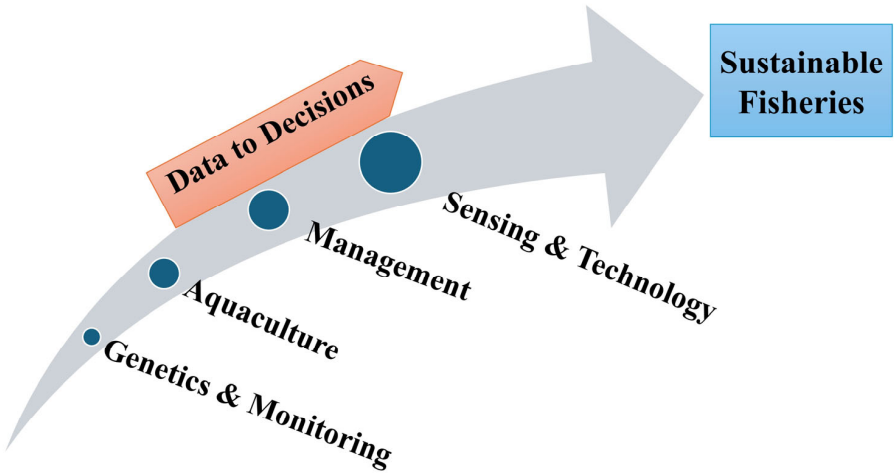

Figure 3. Data to Decisions Framework in Fisheries.

7.1. Data Sources, Benchmarks, and Dataset Bias

The quality, composition, and documentation of datasets are as critical as the selection of model architecture in applications ranging from underwater video and echograms to eDNA and farm telemetry. Extensive, publicly available benchmarks like DeepFish, comprising approximately 40,000 in situ images from 20 tropical marine habitats with classification, localization, and segmentation annotations, facilitate systematic comparisons of detection and segmentation methods in realistic habitat complexity and lighting conditions [11]. Recent corpora, such as the BD freshwater fish dataset, which includes 4389 images of 12 freshwater species from markets in Bangladesh, enhance coverage in small-scale and data-deficient contexts. However, they do so at a significantly smaller scale and with limited taxonomic and environmental diversity [65]. Numerous studies report exceptionally high point accuracies for species recognition, typically 95–99 percent. Still, these results are derived from curated datasets with a limited number of classes, balanced class distributions, and relatively uniform imaging conditions. Consequently, such figures provide only marginal insights into performance in practical applications [5,64]. In contrast, detectors trained and evaluated on more complex datasets that combine various habitats or integrate multiple corpora, such as DeepFish and OzFish, generally exhibit significant declines in precision and recall for small and occluded individuals, rare taxa, or under turbid conditions [32,66]. These patterns indicate that the accuracy of headlines is not significant unless it is explicitly grounded in the relevant data context, which encompasses factors such as the quantity of images or video hours, the number of species or classes, the degree of class imbalance, the presence of long-tail or protected taxa, geographic and seasonal coverage, and the setting of imagery acquisition, whether in laboratories, on farms, or in open water [176,177]. Current aquatic datasets exhibit significant bias towards commercially valuable or easily observable species, temperate regions, and conditions of high visibility. There is inadequate representation of cryptic taxa, deep-water environments, and fisheries characterized by sparse data. Domain adaptation studies consistently indicate that detectors optimized for large-bodied reef fish in clear waters may not generalize effectively to small schooling fish in turbid estuaries or nocturnal settings, despite maintaining the same underlying architecture [11,32,178]. Another issue is the widespread reliance on proprietary or nonstandardized datasets without sufficient transparency: numerous commercial electronic monitoring and farm telemetry implementations depend on private datasets whose sampling design, annotation protocols, and long-term maintenance are not disclosed, hindering independent audits and replication and fostering optimistic reporting [137,179,180]. Current best practices in underwater vision and fisheries acoustics

involve publishing benchmark subsets with comprehensive metadata and well-defined training, validation, and testing splits. Additionally, it is essential to report on the sampling design, labeling quality control, and inter-annotator agreement, while clearly distinguishing between performance on public benchmarks and results derived from proprietary operational data. This approach ensures that methodological advancements can be evaluated transparently and comparably [137,176,181].

7.2. Model Architectures, Training Protocols, and Deployment Trade-Offs

The prevalence of certain architectures, notably YOLO variants for detection and U-Net-style models for segmentation, suggests practical trade-offs in speed, accuracy, and deployment ease rather than inherent superiority. Single-stage detectors, including YOLOv5 through YOLOv10 and their lightweight variants such as LFN YOLO and YOLOv8 SBE, are widely used for real-time inference on standard GPUs and embedded devices while achieving competitive mean average precision on underwater and echogram benchmarks [20,32,158]. The U-Net and DeepLab architectures excel at echogram and phenotype segmentation tasks due to their encoder–decoder architecture with skip connections, which effectively preserve the fine spatial details necessary for accurate length, area, and biomass proxies [35,44]. Many incremental variants of YOLO and U-Net, including Class Activation Map–based YOLO (CAM YOLO), Underwater Distributed Neural Observer (UDINO), and various attention or multi-head extensions, demonstrate improvements of only approximately 0.5 to 2 mean average precision points on individual datasets. Furthermore, their performance under distribution shifts is rarely systematically analyzed, which limits assertions of significant architectural progress. Most studies focused on fisheries document architectures but frequently neglect the training protocol. Key decisions regarding learning rate schedules, batch sizes, optimizers, regularization, augmentation strategies, and stopping rules are often missing. However, evidence from ecology and biodiversity informatics indicates that these factors significantly influence reproducibility and out-of-sample performance, particularly in small or noisy datasets [182]. Underwater detection surveys indicate that performance gains associated with new architectures can often be replicated by enhancing augmentation or more meticulously tuning hyperparameters on the same backbone. This suggests that some reported improvements may stem from optimization rather than an actual increase in model capacity. In the context of edge and AIoT deployment, design considerations go beyond accuracy on a static test set. Factors such as model size, floating-point operations, memory footprint, and cooling and power budgets are critical for assessing the feasibility of inference on battery-powered buoys, camera-equipped nets, or small single-board computers in remote hatcheries. However, numerous studies focus solely on detection accuracy or F1 scores, neglecting latency, power, and maintenance costs, thereby providing an inadequate foundation for engineering decisions [158,161]. Future research should report accuracy and efficiency together, incorporating frame rates within realistic hardware and power constraints, parameter counts, quantization methods, and resilience to sensor dropout and intermittent connectivity. Additionally, it should distinctly delineate the contributions of architecture, training procedures, and evaluation variance.

7.3. System-Level Architectures: Digital Twins, AIoT, and Data Collaboratives

System-level architectures integrating digital twins, Artificial Intelligence of Things (AIoT) platforms, and data collaboratives are frequently regarded as the culmination of digital transformation in aquaculture and fisheries. Empirical research indicates that these initiatives function as ambitious socio-technical programs requiring ongoing design, calibration, and governance, rather than serving as pre-packaged solutions [183,184]. Digital-twin dashboards that integrate growth models, water quality forecasts, and market scenarios

can enhance decision-making regarding feeding, harvest timing, and contingency planning, demonstrating documented benefits in both tank- and cage-based systems [185–187]. Nonetheless, continuous calibration to grading and sensor data, enhancements to the bioenergetic and infrastructure modules, and a clear representation of uncertainty are crucial; failing to do so may result in twins producing simulations that appear accurate yet poorly constrained, particularly during infrequent extremes when management errors carry significant consequences [184,185].

AIoT "smart farm" architectures enhance this framework by incorporating dense sensing, edge computing, and cloud analytics for water quality control, feeding, and health monitoring [60,61,183]. Reviews and bibliometric studies indicate ongoing issues related to interoperability, cybersecurity, ambiguous data ownership, unstable business models for small producers, and a tendency toward short-lived pilots in research-intensive or capital-rich environments rather than the establishment of sustainable infrastructure in remote coastal regions [60,188,189]. Proposals for privacy-preserving data collaboratives, such as federated learning and secure aggregation, offer the potential to jointly train models without sharing raw data. However, these initiatives face challenges, including entrenched power asymmetries, ambiguous contracts, limited farmers' control over secondary use, and governance and funding models that seldom guarantee long-term stability [190,191].

Emerging applications of generative and advanced AI for simulating future climate and disease scenarios in aquatic production indicate that while models in climate and hydrology can replicate complex extremes, they also carry historical biases and often fail to adequately represent compound or unprecedented events unless they are closely guided by physical knowledge and subjected to rigorous evaluation [192]. Current evidence suggests that digital twins, AIoT platforms, and data collaboratives represent system-engineering challenges that integrate sensing, modeling, governance, and finance. Their success relies more on robust standards, enforceable and transparent data governance, sustained maintenance, and participatory design involving farmers, fishers, and regulators, rather than on incremental improvements in neural-network architectures [183,191].

7.4. Metrics, Uncertainty, and Statistical Reliability

A common limitation in the literature is reliance on single-run point estimates, such as overall accuracy, F1 score, or mean average precision, without accounting for variability or uncertainty. This oversight masks the distinction between, for instance, 95 percent accuracy with narrow variation across folds and seeds and 95 percent with wide variation, a difference that carries significant implications for management decisions [193–195]. The issue in fisheries and aquaculture is exacerbated by a common misalignment between machine-learning evaluation metrics and management objectives [196]. High pixelwise Intersection over Union or F1 scores for lesion or object detection in images and video do not ensure precise estimates of disease prevalence, mortality risk, or treatment benefit at the cage or farm level, despite models performing well on standard benchmarks [8,121,197]. Minor improvements in mean average precision may be operationally insignificant compared with systematic biases in abundance indices stemming from class imbalance, long-tail or protected taxa, and domain shift [198,199]. Current evaluation pipelines infrequently incorporate predictive uncertainty into downstream stock assessments, harvest control rules, feeding controllers, or indicators of risk from illegal, unreported, and unregulated fishing. This omission restricts the interpretability of key performance metrics for management decisions [196]. Recent research on robust machine-learning evaluation in ecology emphasizes the need to distinctly separate training, validation, and test datasets using appropriate spatial, temporal, or vessel-level blocking. Additionally, it highlights the importance of reporting complementary diagnostics, including per-class precision–recall

curves, calibration plots, and cost-sensitive losses designed for rare events and skewed class distributions [195,200], Additionally, it highlights the importance of providing uncertainty estimates through methods such as bootstrapping, repeated cross-validation, conformal prediction, or Bayesian approaches when outputs are intended to inform policy or management [182,193,195]. Few studies focused on fisheries meet these criteria, which diminishes the credibility of commonly reported figures claiming 90 percent or higher accuracy or hotspot accuracy. Artificial intelligence systems that inform regulatory or operational decisions, such as environmental DNA-based community indices, automated bycatch estimates from electronic monitoring, tuna biomass estimates under drifting fish aggregating devices, and reinforcement learning controllers for aeration and feeding, must prioritize statistical uncertainty and model drift as key outputs. This necessitates the reporting of prediction intervals for abundance and biomass, clear threshold-dependent trade-offs between false positives and false negatives in risk scores, and safety margins in control policies under plausible worst-case disturbances [194].

7.5. The Operational Gap: From Prototypes to Decisions

Addressing the disparity between data and decision-making remains a core challenge for the advancement of fisheries, artificial intelligence, and automation. Recent research on machine-learning applications and harvest strategy design in fisheries highlights that algorithms are effective only when integrated into transparent decision-making frameworks that include clearly defined objectives, performance metrics, and risk tolerances [196,201,202]. Detection and classification systems currently achieve or surpass human performance in various operational tasks, such as purse seine catch analysis, electronic monitoring, and coastal visual surveys. However, most implementations remain proof-of-concept tools that lack regulatory validation and are not closely integrated with formal harvest strategies [196,203–205]. Validation of decision grades requires more than mere high-point accuracy; it requires a thorough examination of the propagation of predictive uncertainty across stock assessment models, biological reference points, harvest control rules, and socioeconomic indicators. This analysis employs simulation tools, such as management strategy evaluation, reference-point software, and structured decision analysis, to assess whether risks remain within acceptable limits across various scenarios [202,206–209]. Currently, most artificial intelligence studies in fisheries focus on offline evaluations of prototype algorithms. There is minimal engagement with management strategy evaluations or similar closed-loop frameworks to assess performance in the context of realistic feedback, observation error, implementation uncertainty, and stakeholder constraints [196,210]. Future research should prioritize the development of fully integrated pipelines that process raw sensor data to produce calibrated decision alternatives, including associated prediction intervals and explicit risk trade-offs. This is particularly relevant for automated feeding rules in aquaculture or for dynamic spatial closures in wild-capture fisheries. Additionally, these pipelines should undergo in silico evaluation through management strategy assessment before operational deployment [201,202,207,208]. The transition from standalone pattern recognition to rigorously tested decision support systems is essential; otherwise, artificial intelligence will continue to function merely as a diagnostic supplement rather than as a reliable element in the precautionary ecosystem-based management of aquatic resources [196].

8. Practical Implications

For policymakers and regulators, fusion-based risk scoring (AIS/VMS with SAR/VIIRS), mandated uncertainty reporting for eDNA, acoustics, and EM, and support for open, versioned benchmarks can align AI with transparent, defensible enforcement. Fisheries

managers and Regional Fisheries Management Organizations (RFMOs) should incorporate uncertainty-aware indices into harvest control rules, pair dark fleet detection with targeted patrols, and evaluate deterrence and compliance gains relative to cost. Aquaculture operators can deploy precision feeding and DO/pH/NH_3 forecasting with drift monitoring and confidence-aware throttling, embed digital-twin planning for harvest and contingencies, and report welfare and environmental KPIs alongside productivity. While federated learning and differential privacy offer theoretical solutions to data sharing, their practical implementation faces technical hurdles (communication overhead, accuracy loss) and institutional barriers (liability, competitive concerns) that remain largely unresolved. For researchers, priorities include rare-taxa corpora, physics- and trait-aware augmentation, conformal or Bayesian uncertainty tooling integrated into decisions, and manager-in-the-loop trials that quantify real operational value. Collectively, these steps can move AI from promising prototypes to operational infrastructure that advances sustainability, resilience, and equity across both industrial and small-scale fisheries.

9. Conclusions

This review synthesizes recent advances in artificial intelligence applicable to fisheries and aquaculture, highlighting the significant potential of computer vision, acoustic inference, genomic analysis, and Artificial Intelligence of Things architectures when integrated into well-designed systems. This document outlines enhancing species and trait detection through AI-driven monitoring, improving production efficiency and animal welfare via precision aquaculture tools, and strengthening monitoring, control, and surveillance through multi-sensor fusion. Ongoing system-level limitations hinder the transition from laboratory prototypes to large-scale operational infrastructure. The robustness of models across various domains is inadequate, as trained systems frequently do not generalize well across different habitats, seasons, fleets, and gear types, especially for rare and protected taxa. Additionally, the availability of standardized multi-region datasets is limited. Uncertainty is infrequently transmitted from model outputs to assessment and management endpoints, leaving decision-makers challenged to evaluate the reliability limits of AI-assisted recommendations. Governance and institutional arrangements are not keeping pace with technical innovation: interoperability among vendors is incomplete, cybersecurity and data privacy protections are inconsistent, and numerous small-scale operators lack access to data infrastructure and influence over governance processes.

In this context, it is improbable that the field will transform solely through incremental improvements in accuracy from novel model architectures. Significant advancements will arise from prospective, multi-site validation in actual fisheries and aquaculture settings; from open, versioned benchmark datasets encompassing rare, cryptic, and geographically diverse taxa that facilitate rigorous out-of-sample testing; and from deployment frameworks that address temporal drift, quantify comprehensive life cycle costs, and connect algorithmic outputs to measurable enhancements in stock status, environmental impact, and the livelihoods of fishers and farmers. By prioritizing these considerations, artificial intelligence can transition from a cutting-edge technology used in limited pilot projects to a reliable and equitable component of the global infrastructure supporting sustainable fisheries and aquaculture.

Table 1. AI Applications in Fisheries Genetics and Monitoring: Methods, Efficacy, Limitations, Validation Context, and Data Maturity.

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
Species Recognition (Underwater)	CNNs (ResNet, YOLO, EfficientNet) on underwater imagery	>95% accuracy; real-time inference	High accuracy (>95%) from curated datasets (~500–600 images) not representative of operational data (rare taxa, variable lighting). Domain shift: severe degradation under turbidity. Propagation to abundance indices undocumented.	Mixed (lab-dominant)	Experimental	[11,12,50]
Transfer Learning	Fine-tuned CNNs with domain-specific retraining	Reduces annotation burden 50–80%	Domain-specific biases degrade rare-species recall. Requires >100 labeled images per species. Success depends on source-target similarity.	Experimental	Prototype-Validated	[14,67,211]
Image Restoration	Super-resolution (ESRGAN) + task-aware CNNs	Restores detail; <5% false positives	Hallucination risk distorts ecological endpoints (counts/lengths). Uncertainty quantification rarely implemented. Field validation limited.	Lab-validated	Experimental	[83,84]
Temporal Sequence Modeling	CNN + RNN/LSTM/GRU on video streams	Improves video recognition; enables behavior classification	Requires 100 s–1000 s video hours. Overfitting to single-farm telemetry common. Cross-validation rigor inconsistent.	Lab-limited	Experimental	[64,212,213]
Morphometrics & Biomass	Stereo vision + regression (RF, NN)	Non-invasive length/biomass estimates	Sensitive to water clarity; accuracy degrades with turbidity. Error propagation to stock assessment not quantified.	Lab-validated	Prototype	[212,214–216]
Behavioral Monitoring	Computer vision tracking (multi-object, pose estimation)	Non-invasive stress detection; real-time alerts	Computationally heavy. Poor cross-farm generalization (lighting, density). Label scarcity for specific behaviors.	Lab-to-field	Experimental	[117,120]
Environmental DNA	ML classifiers on barcode data; taxonomic assignment	Non-invasive community detection; aligns with traditional indices	Reference library bias for understudied taxa. Read counts vs. abundance correlation unresolved. Standardized controls inconsistently reported.	Validated (field + lab)	Validated	[3,88,217]
Acoustic Species Inference	Deep learning on echograms (CNN, weakly supervised)	Non-visual ID; survey-grade potential; continuous monitoring	Severe data scarcity (~10 s public corpora globally). No cross-frequency standardization. Proprietary systems limit reproducibility.	Prototype	Prototype	[40,52,218]
Multi-Sensor Fusion	Imaging + eDNA + acoustics + trait modeling	Ecosystem-scale monitoring; reduced single-modality bias	High integration complexity. Data governance standards absent. Theoretical in most contexts. Long-term wild validation lacking.	Experimental	Experimental	[89,90,219]

Table 2. AI Applications in Aquaculture: Methods, Efficacy, Limitations, Validation Context, and Data Maturity.

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
Disease Detection	CNNs, region-based CNNs, spectral/thermal imaging	>90% accuracy; non-invasive lesion identification	High accuracy (>90%) from controlled tanks not representative of crowded commercial environments. False positives/negatives high under variable lighting/turbidity. Dataset bias toward visible lesions; cryptic infections missed. Intervention efficacy (mortality reduction) undocumented	Lab-dominated	Experimental	[111,121,128,220]
Feeding Optimization	Gradient boosting, SVR, Deep Deterministic Policy Gradient (RL)	Reduces waste 20–30%; improves FCR	Farm-specific calibration (>50 h) required. Overfitting to single-farm telemetry common. Models fail with sensor/feed changes or density variation. Generalization poor; decision thresholds opaque	Lab-to-field (mixed)	Experimental	[25,106,112]
Water Quality Forecasting	IoT arrays + CNN-LSTM / Bidirectional RNN	Predicts DO/pH/ammonia hours in advance	Sensor fouling and concept drift (biomass/temp changes) degrade long-term accuracy. Single-farm training limits generalization. Forecast uncertainty rarely quantified for control logic	Lab-to-field (prototype)	Experimental	[27,115,221]
Welfare Monitoring	Computer vision tracking (multi-object, pose estimation)	Non-invasive stress detection; real-time alerts	Poor visibility in crowded tanks limits utility. Generalization across farm types (lighting/tank config) weak. Label scarcity for behavioral events. Intervention validation absent	Lab-to-field (mixed)	Experimental	[26,117,120]
Harvest Forecasting	Time-series regression, ensemble ML (RF, GBM)	Optimizes harvest timing; synchronizes logistics	Dependent on consistent inputs; degrades with equipment changes. Predominantly retrospective; prospective validation rare. Decision uncertainty (confidence intervals) seldom quantified	Lab-limited	Experimental	[6,122,222,223]
Climate Resilience	Generative AI (diffusion, transformers) + physics-informed models	Scenario planning for heat/hypoxia/pathogens	Experimental; lack of empirical validation. Risk of hallucinating biologically implausible scenarios. Biased prior assumptions. Calibration protocols absent	Experimental	Prototype	[126,130,224,225]
Omics-Integrated Feeding	Transcriptomics/metabolomics + ML classifiers	Phenotype-specific precision nutrition	Large multi-omics datasets impractical for commercial farms. Causality between omics signatures and growth response unclear. Cost–benefit analysis absent	Lab-only	Prototype	[226]
Biofloc Optimization	IoT + ML (Random Forest, SVM) for TSS/ammonia prediction	Automates C/N ratio maintenance; reduces operator burden; stabilizes system	High microbial variability across carbon sources (molasses vs. starch) limits model transferability. TSS sensors prone to fouling. Cross-system generalization (shrimp→fish) poor. Bacterial abundance classification undocumented	Lab-to-Field (mixed)	Experimental	[227–230]
AIoT Smart Farming	Edge-cloud IoT + ML retraining; automated control	Integrated real-time farm automation; labor reduction	High cost; internet reliance problematic in remote regions. Interoperability/security standards absent. ROI for smallholders unclear. Lifecycle maintenance unquantified	Prototype (pilot)	Experimental	[23,60,61]

Table 3. AI Applications in Fisheries Management: Methods, Efficacy, Critical Limitations, Validation Context, and Data Maturity.

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
IUU Detection	AIS/VMS + anomaly detection (CNN, RNN, Gradient Boosting)	>90% hotspot accuracy; identifies loitering/transshipment	False positives high in crowded zones. Dependence on voluntary AIS transmission enables spoofing/gaps. Definition of "illegal" varies by jurisdiction; legal validation difficult. Systematic exclusion of small (<12 m) vessels	Field (global)	Validated	[29,30,134,231]
Fishing Effort Mapping	Satellite AIS + CNN gear classifiers; SAR fusion	Identifies global fishing footprint; tracks industrial fleets	Systematic exclusion of artisanal/dark fleets (non-AIS). Cloud cover limits optical validation. Resolution limits (>10–20 m vessels only). Historical baselines biased by increasing AIS adoption rates (not just effort increase)	Field (global)	Validated	[2,57]
Electronic Monitoring (EM)	EM video + CV classifiers (YOLO, ResNet)	Automates catch/bycatch logs; reduces observer cost/risk	High hardware cost; massive data storage requirements. Human-in-the-loop required for complex scenarios. Performance degrades with lens fouling, water spots, night lighting. Privacy concerns limit acceptance	Field (pilot/regional)	Experimental	[8,135,137]
Protected Species Tracking	EM + AI classification (fine-grained recognition)	Improves endangered species monitoring; precise bycatch quantification	Rare events cause extreme class imbalance (few training examples). Taxonomic resolution limited (e.g., "shark" vs. "silky shark"). Expert validation required for legal enforcement. Detection probability varies by camera angle	Field (mixed)	Experimental	[141,142,144,232]
Dark Fleet Detection	VIIRS (night lights) + SAR + ML fusion	Reveals non-AIS vessels; independent of cooperative tracking	High cost of commercial SAR data; revisit times create temporal gaps. Sensitive to cloud cover (optical) and sea state (SAR clutter). Matching dark targets to vessel identity remains challenging without visual confirmation	Field (regional)	Validated	[31,58,146]
Transshipment Detection	AIS clustering + ML (loitering algorithms)	Identifies at-sea transfers (IUU/labor risk); supply chain transparency	False positives from legitimate rendezvous. Behavior-based only (cargo transfer unconfirmed without visual). "Dark" transshipments (AIS off) missed without SAR fusion. Validated primarily for large carrier vessels	Field (global)	Validated	[149,150]

Table 3. *Cont.*

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
Optical-AIS Fusion	Deep Learning on optical imagery + AIS correlation	Detects small, untracked vessels; verifies AIS position data	Severely cloud dependent. Daytime operation only. High data latency; computationally expensive for global scan. Resolution limits detection of small artisanal boats (<10 m)	Prototype	Experimental	[152,155]
EEZ Monitoring	EO (Earth Observation) + AIS + ML geofencing	Detects unauthorized EEZ incursions; sovereign protection	High operational costs for continuous monitoring. Legal admissibility of satellite evidence varies globally. Real-time alerting latency (satellite downlink delays) limits interception	Field (regional)	Experimental	[153,154]
Transparency Frameworks	Big data synthesis + ML visualization	Guides governance and oversight; public accountability	Implementation gap: data exists but enforcement lacks resources. Global North bias in monitoring capability. Sovereignty concerns regarding public data sharing. Standardization of "transparency" metrics absent	Policy/ Framework	Validated	[31,180]

Table 4. AI Applications in Fisheries Technology: Methods, Efficacy, Limitations, Validation Context, and Data Maturity.

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
Active Acoustic Classification	Deep learning (CNN, weakly supervised) on multi-frequency echograms	Automates schooling species identification; reduces expert annotation effort	Annotated data scarcity (the "annotation bottleneck"). Classification drift due to behavioral/seasonal morphology changes. Reproducibility hindered by proprietary echo-sounder formats	Field (commercial)	Experimental	[40,52,89]
Passive Acoustic Monitoring (PAM)	CNNs, Random Forests on hydrophone audio; source separation	Non-invasive monitoring of spawning events; species-agnostic potential in dark/turbid water	Restricted to soniferous species (many commercial fish silent). High ambient noise (vessels, wind) degrades signal-to-noise ratio. Cryptic vocal repertoires remain undocumented	Field (mixed)	Experimental	[162–164]
Real-Time Biomass Estimation (Buoys)	Echo-sounder buoys + Gradient Boosting/RF (TUN-AI)	Fishery-independent abundance indices (e.g., FAD-associated tuna); optimizes fleet logistics	Reliance on proprietary "black-box" algorithms. Static Target Strength (TS) assumptions ignore size/orientation variance. Discrimination of similar species (e.g., Katsuwonus pelamisvs.Thunnus albacares) remains error-prone	Field (operational)	Validated	[39,42]

Table 4. *Cont.*

Feature	Methods	Benefits	Limitations	Validation Context	Data Maturity	References
Echogram Object Detection	YOLO (v5/v8), lightweight CNNs on sonar imagery	High-precision detection in clutter; enables real-time edge computing	Performance degrades with dense/occluded schools. Bounding box precision inferior to segmentation for amorphous shapes. Poor generalization across frequency bands	Lab-to-Field	Experimental	[156,158]
Autonomous Acoustic Surveys	Unmanned Surface Vehicles (USV) + onboard ML	Continuous low-cost monitoring; access to shallow/hazardous zones	Energy constraints limit real-time ML processing. Fish avoidance behavior to USV noise/shadow poorly characterized. Sea state limits operational windows	Field (pilot)	Experimental	[59,160]
Fishing Ground Prediction	U-Net, Deep Learning on oceanographic data (SST, Chl-a)	Predicts high-catch zones; reduces search time and bycatch risk	High sensitivity to climate regime shifts (training data obsolescence). Spatial resolution mismatch between satellite data and local features. Risk of accelerating over-exploitation	Field (hindcast)	Validated	[6,233]
Target Strength (TS) Calibration	In situ measurements + 3D modeling + ML regression	Improves biomass accuracy; accounts for physiological variance	Species-specific and costly to acquire. Depth-dependent swimbladder compression alters TS nonlinearly. Ex situ (lab) measurements often diverge from wild behavior	Lab/Field (mixed)	Experimental	[43]
Hybrid Multi-View Fusion	Fusion of acoustics + optical/environmental data	Leverages complementary signals to improve classification confidence	High computational complexity. Synchronizing asynchronous data streams is technically challenging. Alignment of acoustic/optical fields-of-view difficult in dynamic water	Prototype	Experimental	[145,234,235]

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